STATISTICS · UNIT 4: STATISTICAL INFERENCE

The Later Start Investigation

Use confidence intervals and significance tests to advise the school board on a later start time.

CP1

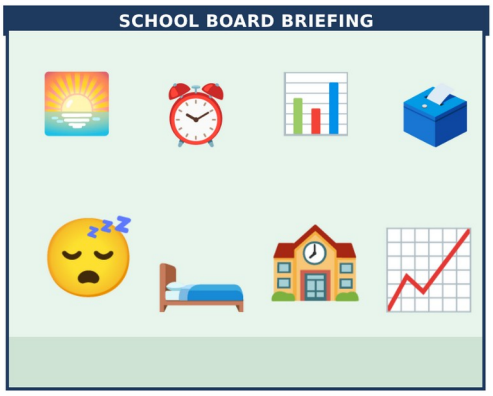
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The school board is considering starting school 45 minutes later. Board members want to know whether most students support it and whether students are really short on sleep.

Your team will collect evidence with confidence intervals and significance tests and present a recommendation with the right amount of certainty.



Survey

248 of 400 support

$\hat{p} = 0.62$

Sleep

16 students

$\bar{x} = 6.8$ h, $s = 1.2$ h

Learning Goals

- I can construct and interpret z-intervals for a proportion and t-intervals for a mean.
- I can carry out one-sample z- and t-tests with conditions.
- I can explain Type I and Type II errors and choose a sample size.

Project Timeline

When	What you will do
Day 1	Estimate Support
Day 2	Are Students Short on Sleep?
Day 3	Errors and Grades
Final	Report to the School Board + reflection

Part 1: Estimate Support

Day 1 · Use the Support survey card (SRS of 400, 248 support).

Evidence Cards

Support survey	By grade	Sleep sample
SRS of 400 students	Grade 9: 130 of 200	SRS of 16 students
248 support the change	Grade 12: 118 of 200	$\bar{x}$ = 6.8 h, s = 1.2 h

Doctors say	Critical values	Pilot week
Teens need about	z*: 1.645 (90%), 1.96 (95%)	10 students: extra sleep (h)
8 hours a night	2.576 (99%)	0.5, 1, 0.25, 0.75, −0.25, 1.25, 0.5, 0, 1, 0.5

1. Check the conditions (Random, 10%, Large Counts) for a one-proportion z-interval.

2. Construct and interpret a 95% confidence interval.

3. How large a sample is needed for a margin of error of at most 3% at 95% confidence (use $\hat{p} = 0.5$)?

4. Test H_0: $p = 0.5$ vs H_a: $p > 0.5$ at $\alpha = 0.05$. Give z, the p-value, and a conclusion.

Part 2: Are Students Short on Sleep?

Day 2 · Use the Sleep sample card (n = 16, $\bar{x}$ = 6.8 h, s = 1.2 h).

5. Construct a 95% t-interval for the mean sleep of all students (df = 15, $t^* \approx 2.131$).

6. Test H_0: $\mu = 8$ vs H_a: $\mu < 8$. Find t and the p-value.

7. Write a conclusion in context. What condition about the data must you check?

Part 3: Errors and Grades

Day 3 · Use the By grade card.

8. Describe a Type I and a Type II error for the sleep test.

9. Find $\hat{p}$ for each grade and the difference $\hat{p}_9 - \hat{p}_{12}$.

10. Construct a 95% confidence interval for $p_9 - p_{12}$ and interpret it.

11. How would increasing α from 0.05 to 0.10 affect the chances of each error?

Final Product: Report to the School Board

Your final product must include:

- One-proportion interval, sample size, and test
- Mean sleep interval and t-test
- Errors in context
- Comparison of grades
- Report to the school board

Reflection

R1. Why do we use t instead of z for the sleep data?

R2. Why does the grade comparison not prove the grades feel the same?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Intervals and tests are chosen, carried out, and interpreted correctly and support the recommendation. Goes beyond the minimum with a creative, realistic choice.	Intervals and tests are chosen, carried out, and interpreted correctly and support the recommendation.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report states the evidence, uncertainty, and limits of the study clearly for a non-statistician. Polished, organized, and visually engaging.	Report states the evidence, uncertainty, and limits of the study clearly for a non-statistician. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.